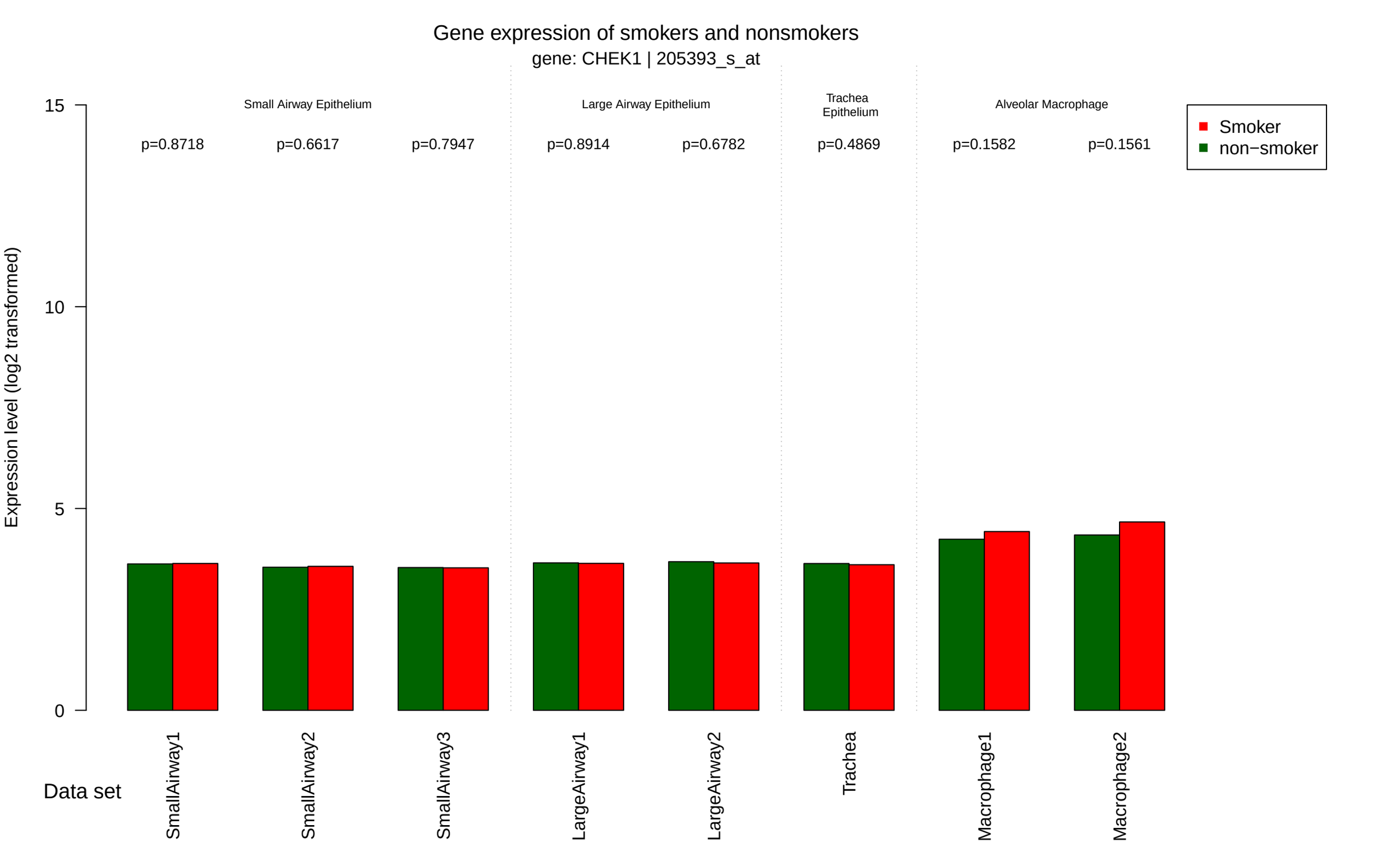
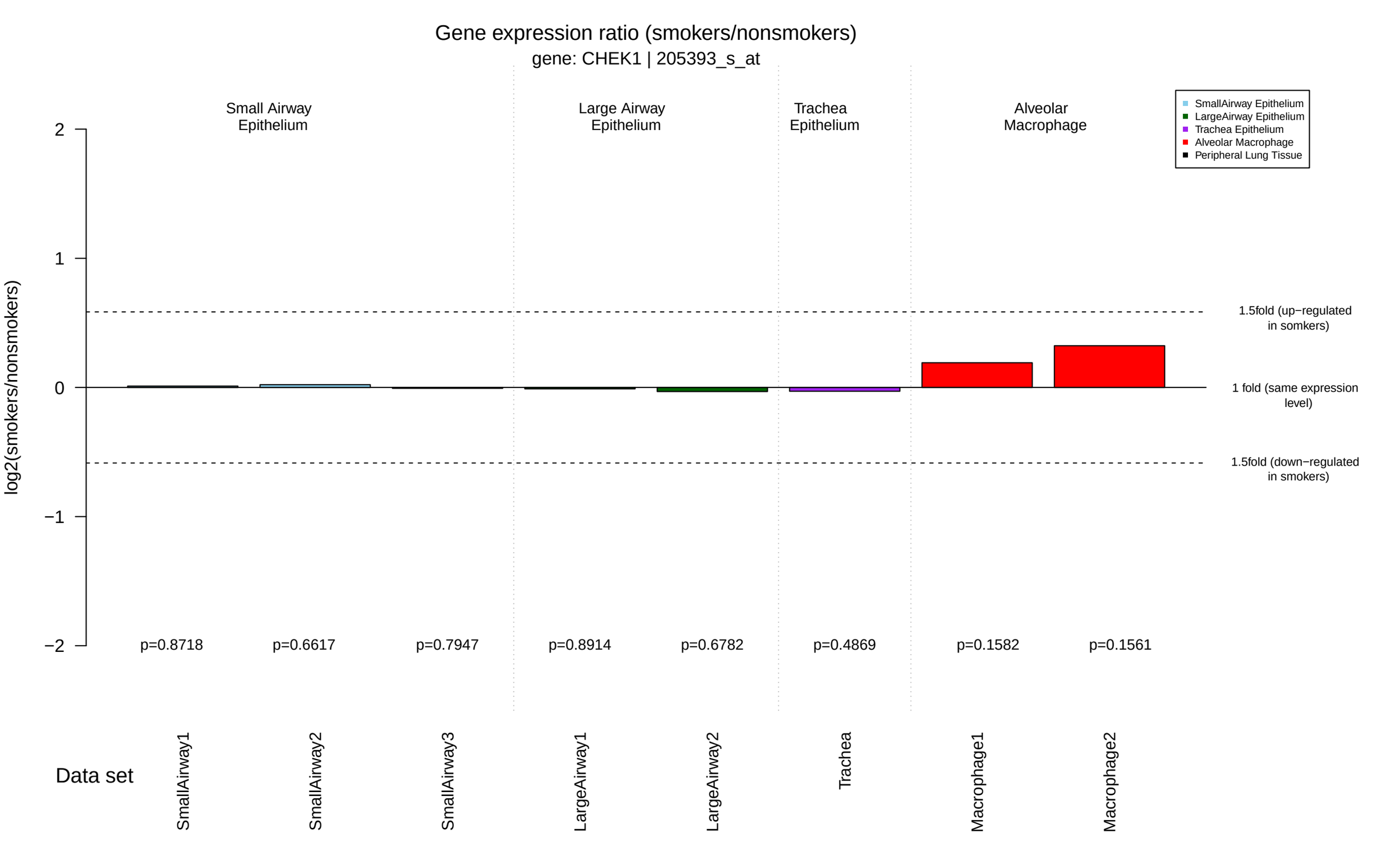
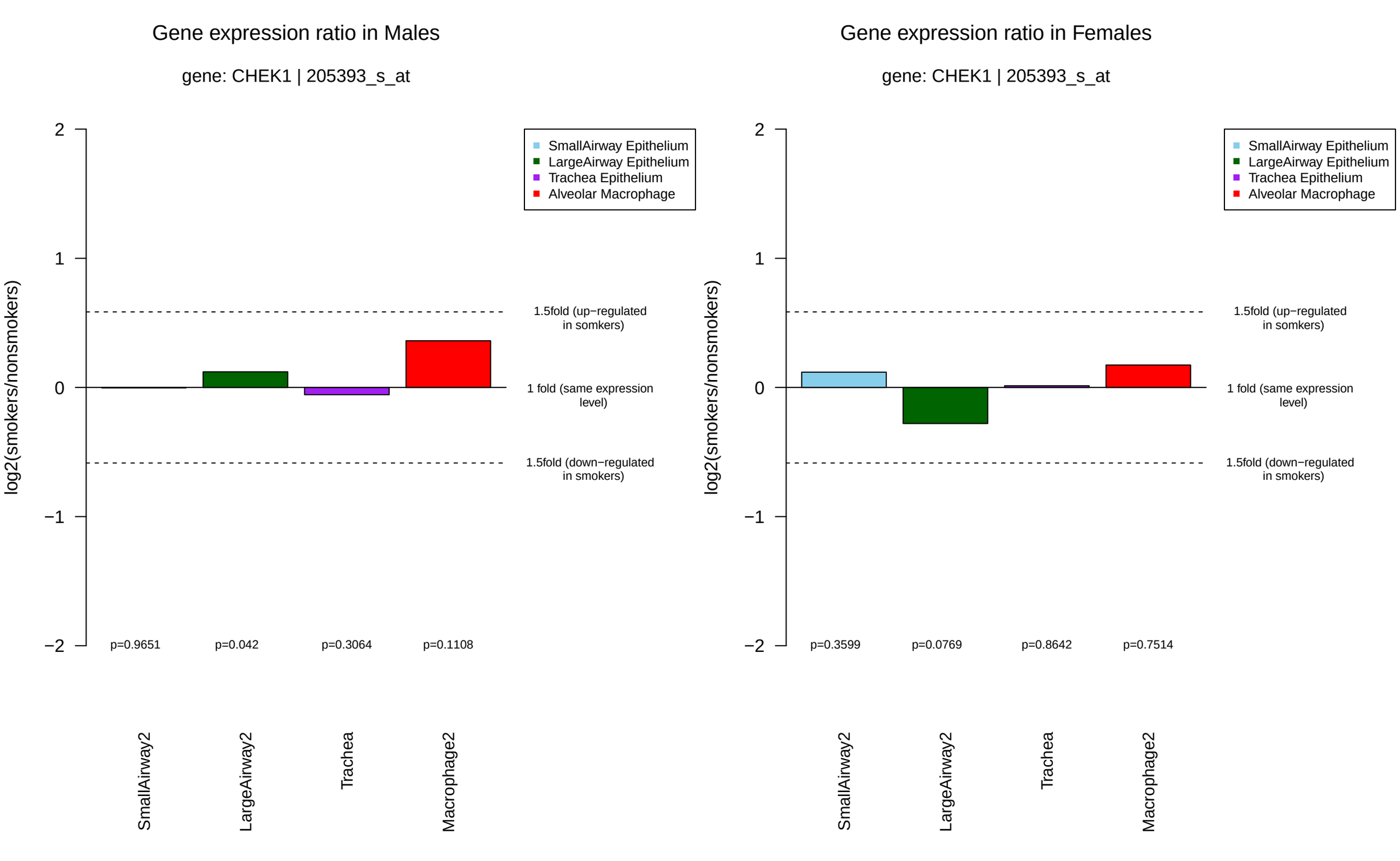
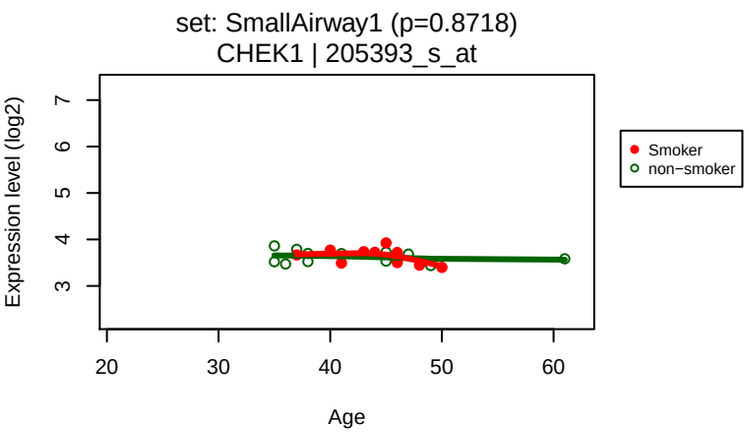
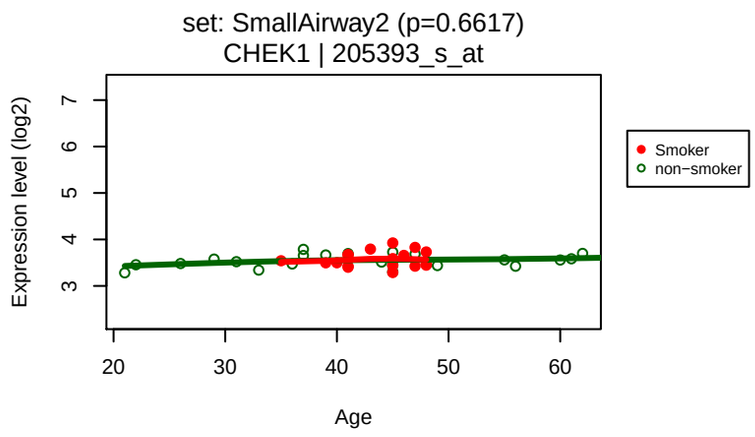
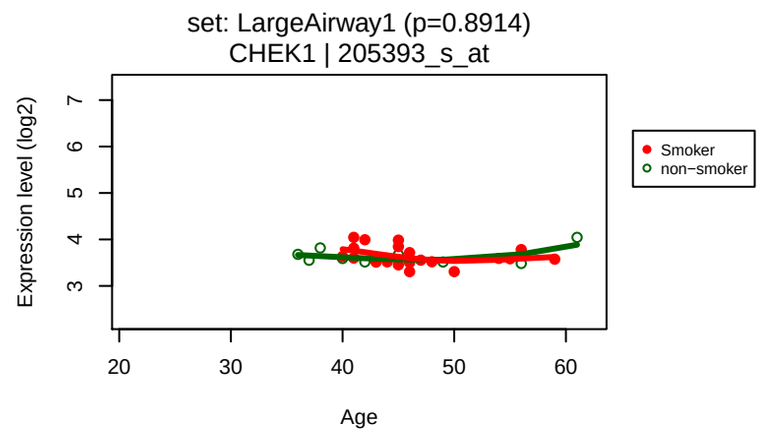

Gene expression of smokers and nonsmokers
gene: CHEK1 | 205393_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8718
p=0.6617
p=0.7947
p=0.8914
p=0.6782
p=0.4869
p=0.1582
p=0.1561
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

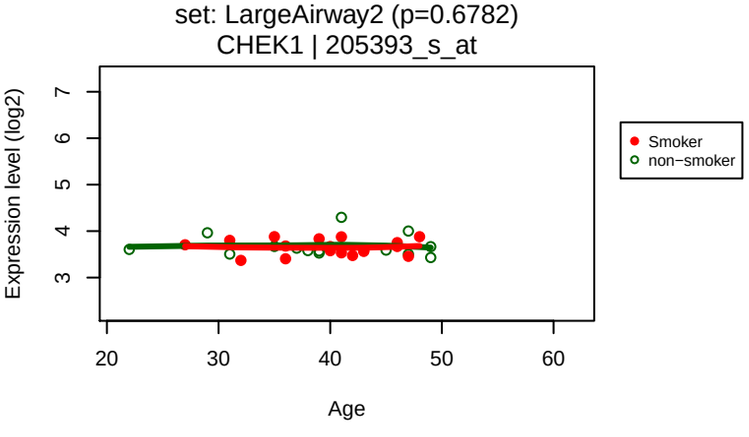

Gene expression ratio (smokers/nonsmokers)
gene: CHEK1 | 205393_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8718
p=0.6617
p=0.7947
p=0.8914
p=0.6782
p=0.4869
p=0.1582
p=0.1561
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

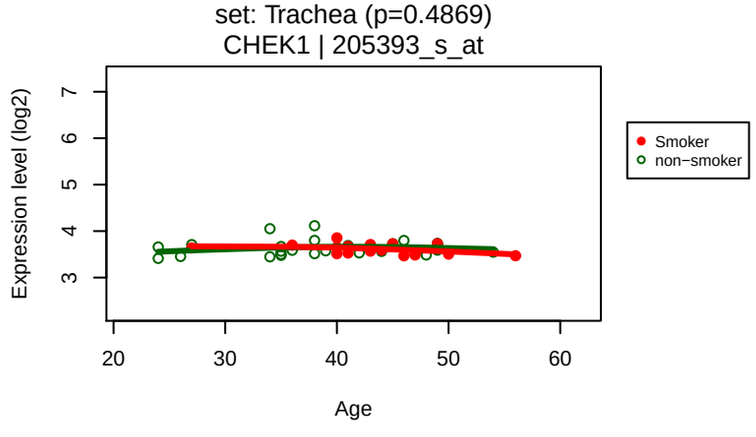

Gene expression ratio in Males
gene: CHEK1 | 205393_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9651
p=0.042
p=0.3064
p=0.1108
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHEK1 | 205393_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3599
p=0.0769
p=0.8642
p=0.7514
SmallAirway2
LargeAirway2
Trachea
Macrophage2

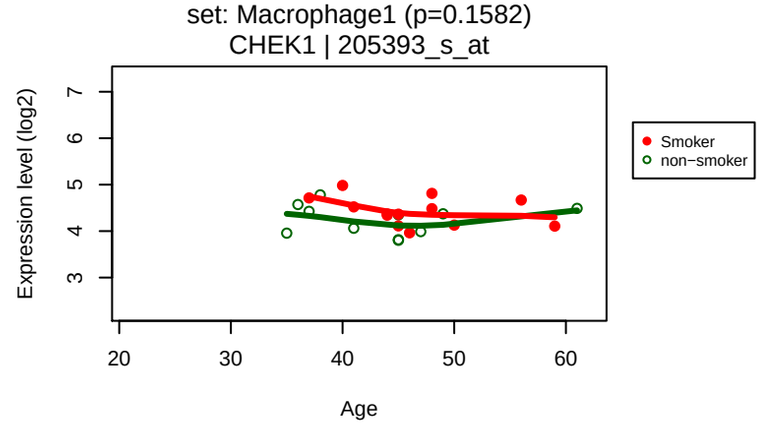

set: SmallAirway1 (p=0.8718)
CHEK1 | 205393_s_at
Expression level (log2)
Age
Smoker
non−smoker

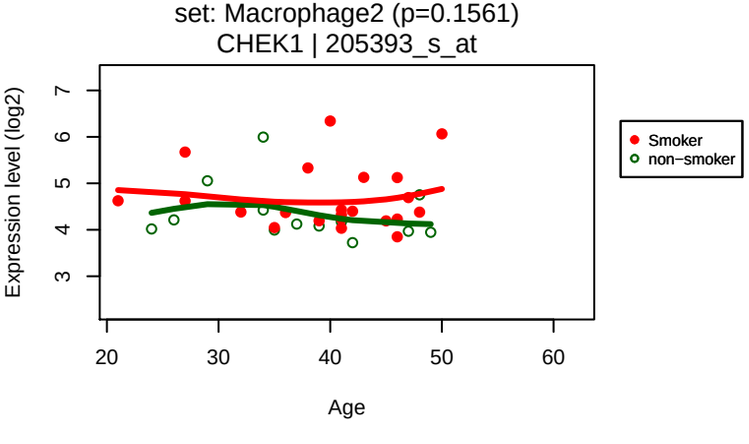

set: SmallAirway2 (p=0.6617)
CHEK1 | 205393_s_at
Expression level (log2)
Age
Smoker
non−smoker

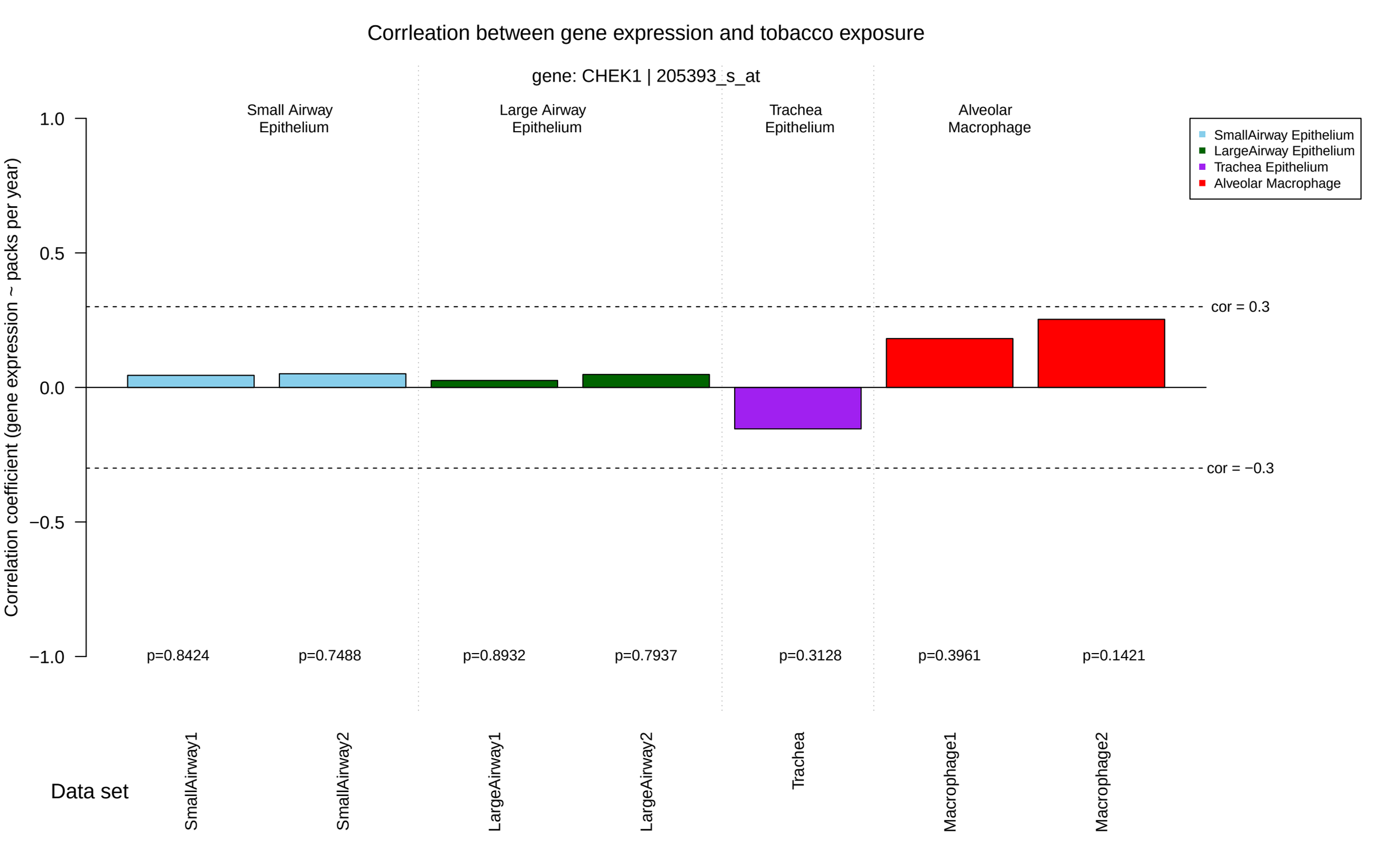

set: LargeAirway1 (p=0.8914)
CHEK1 | 205393_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6782)
CHEK1 | 205393_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4869)
CHEK1 | 205393_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1582)
CHEK1 | 205393_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1561)
CHEK1 | 205393_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHEK1 | 205393_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8424
p=0.7488
p=0.8932
p=0.7937
p=0.3128
p=0.3961
p=0.1421
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2